

Introduction

Sentiment analysis has traditionally relied on text, speech, and visual data, leaving body movement as a comparatively unexplored modality despite its strong link to affective expression. Radar sensing offers a non-intrusive, privacy-preserving alternative capable of capturing subtle body movements without the limitations of cameras or wearables.

This work uses the multimodal RF-Behavior dataset to investigate radar-based sentiment recognition through three complementary approaches shown below.

- 1 Extracting local features and using classical Machine Learning techniques.
- 2 Training on raw multi-modal data using Deep Learning Transformer encoder
- 3 Utilizing Neural Controlled Differential Equation for interpolation and classification

Dataset

The study is built on the third campaign of the RF-Behavior dataset [?], collected from 44 participants using 13 radars, 8 RFID tags, LoRa devices, and infrared cameras. This campaign targets six sentiment states: depression, distraction, excitement, focus, relaxation, and stress.

Radar data consists of XYZ target-position sequences over time; infrared data adds 3–6 body-part landmarks (position and orientation) sampled at 100 Hz. Prior to modeling, radar streams are time-synchronized across sensors and normalized per user to remove individual biases.

Method 1: Feature-Based Classification

Radar trajectories were segmented into 90-frame windows with a 20-frame stride, then reduced to a set of kinematic, statistical, and directional descriptors (speed, acceleration, jerk statistics, spectral measures, tortuosity, turning-angle entropy, PCA-based directionality, etc.), summarized in Table 1.

Category	Examples
Motion dynamics	Speed, acceleration, jerk (mean, std, median, quartiles)
Statistical	Speed CV, skewness, autocorrelation, spectral entropy
Directional	Turn angle, tortuosity, PCA explained variance

Table 1: Extracted trajectory features (27 per radar).

With 13 radars, this yields 351 candidate features across over 600 windows. An ANOVA F-test selects the top-k features per radar, and Leave-One-Subject-Out (LOSO) cross-validation guides a joint search over window size, stride, and feature count for five classifiers: Support Vector Machine (SVM), LightGBM, XGBoost, CatBoost, and MLP.

Results

SVM achieved the strongest performance with weighted F1-score of 0.72, outperforming the gradient-boosting models (LightGBM and XGBoost around 0.66–0.67) and CatBoost (0.62). MLP performed comparably to the boosting methods. These results suggest that carefully engineered motion descriptors can provide strong discriminative power without requiring deep architectures.

Method 2: Deep Learning with Raw Data

This approach trains transformer encoders directly on raw radar and infrared streams from all three RF-Behavior campaigns. Radar point clouds are resampled to a data-driven optimal rate of 18.7 FPS, balancing temporal frame count against spatial point density per frame. The architecture (Figure 1) uses a Point Cloud Encoder (a BERT-style self-attention block, without positional encoding for radar) to condense spatial points into a single per-frame vector, followed by a Temporal Encoder (a LLaMA-style causal decoder with rotary embeddings) that condenses the frame sequence into a single recording-level embedding. For infrared data, an optional convolutional downsampling block can further compress the temporal axis.

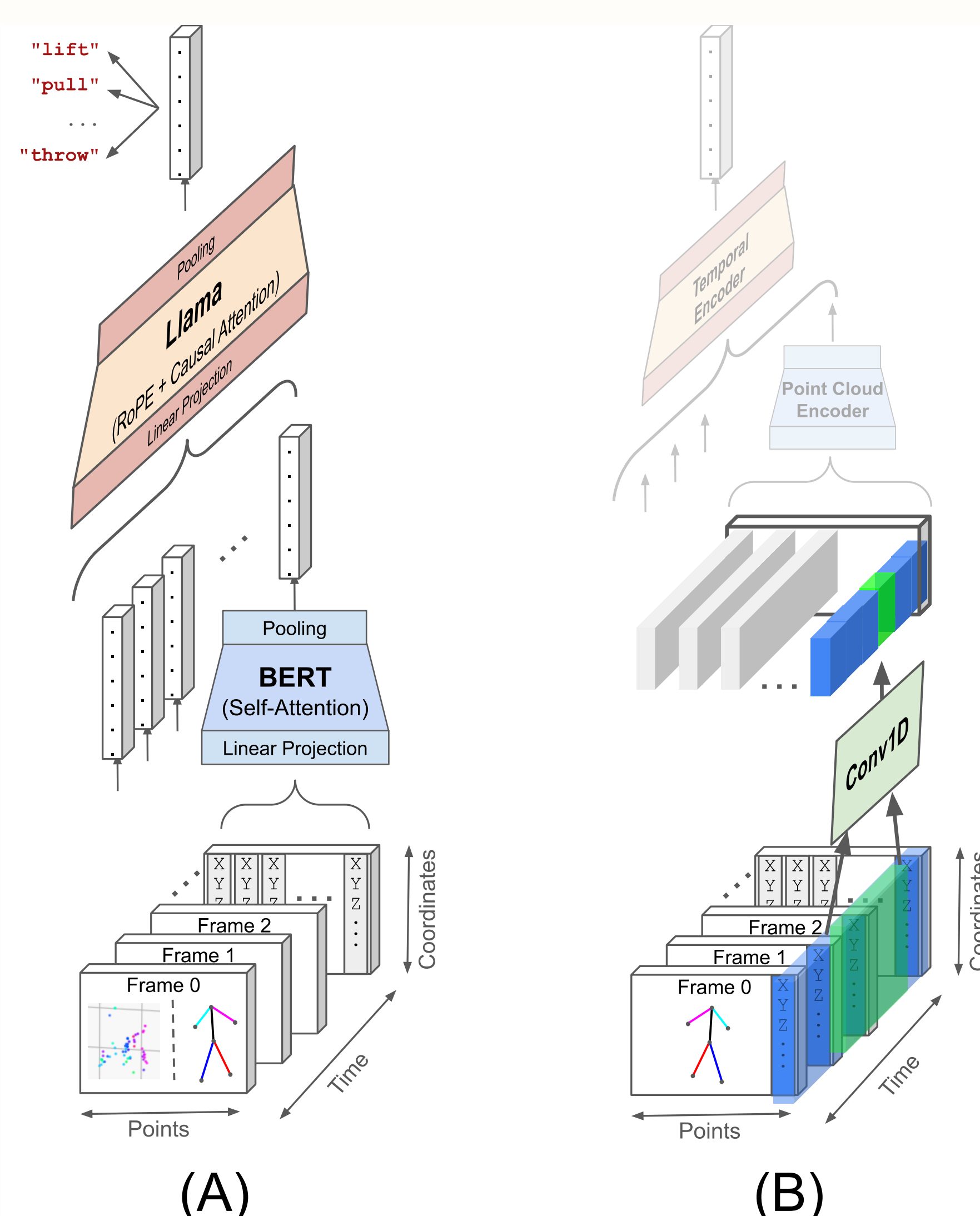


Figure 1: Dual Transformer pipeline

Both supervised classification (cross-entropy loss, user-level split) and unsupervised contrastive learning (NT-Xent loss, class-level split) were evaluated, along with a hyperparameter search over convolutional depth, transformer depth, attention heads, embedding size, and pooling strategy.

Results

Campaign	Classes	Type	Params	Test-F1	Models
C1	21	Radar	1,311,125	94.41%	12
C1	21	IR	526,869	96.56%	25
C2	10	Radar	106,826	91.97%	23
C2	10	IR	1,541,850	98.12%	29
C3	6	Radar	1,349,766	46.12%	30
C3	6	IR	243,798	56.16%	56
C1+C2	31	Radar	120,895	89.51%	6
C1+C2	31	IR	195,375	91.52%	5
C1+C2+C3	37	Radar	268,453	83.74%	5
C1+C2+C3	37	IR	165,269	78.29%	7

Table 2: Results for supervised classification models.

Campaign	Split	Type	Params	Test-F1	Models
C1	17-2-2	Radar	254,144	93.03%	6
C1	17-2-2	IR	142,016	96.68%	7
C2	6-2-2	Radar	93,440	77.51%	14
C2	6-2-2	IR	923,280	93.48%	31
C1+C2	23-4-4	Radar	175,776	90.31%	7
C1+C2	23-4-4	IR	282,176	88.13%	7
C1+C2+C3	25-6-6	Radar	817,472	79.20%	7
C1+C2+C3	25-6-6	IR	85,184	75.41%	7

Table 3: Results for one-shot unsupervised classification. (Split: # train-valid-test behavior classes)

Method 3: Neural Interpolation & Classification

Due to the irregular, asynchronous nature of sensor observations, we attempt interpolation using Deep Learning techniques by employing a Neural Controlled Differential Equation (NCDE) to interpolate and classify the sequences. This method is compared against 1D U-net and Temporal Convolutional Network (TCN) and cubic spline which serves as a baseline.

The workflow is as follows:

- 1 Randomly masked observations serve as interpolation target.
- 2 NCDE requires continuous input, done using hermite cubic spline interpolation.
- 3 NCDE has continuous hidden state, allowing retrieval at any point.
- 4 hidden state is fed into an MLP to obtain predicted value at timestamp, another to classify sequence.

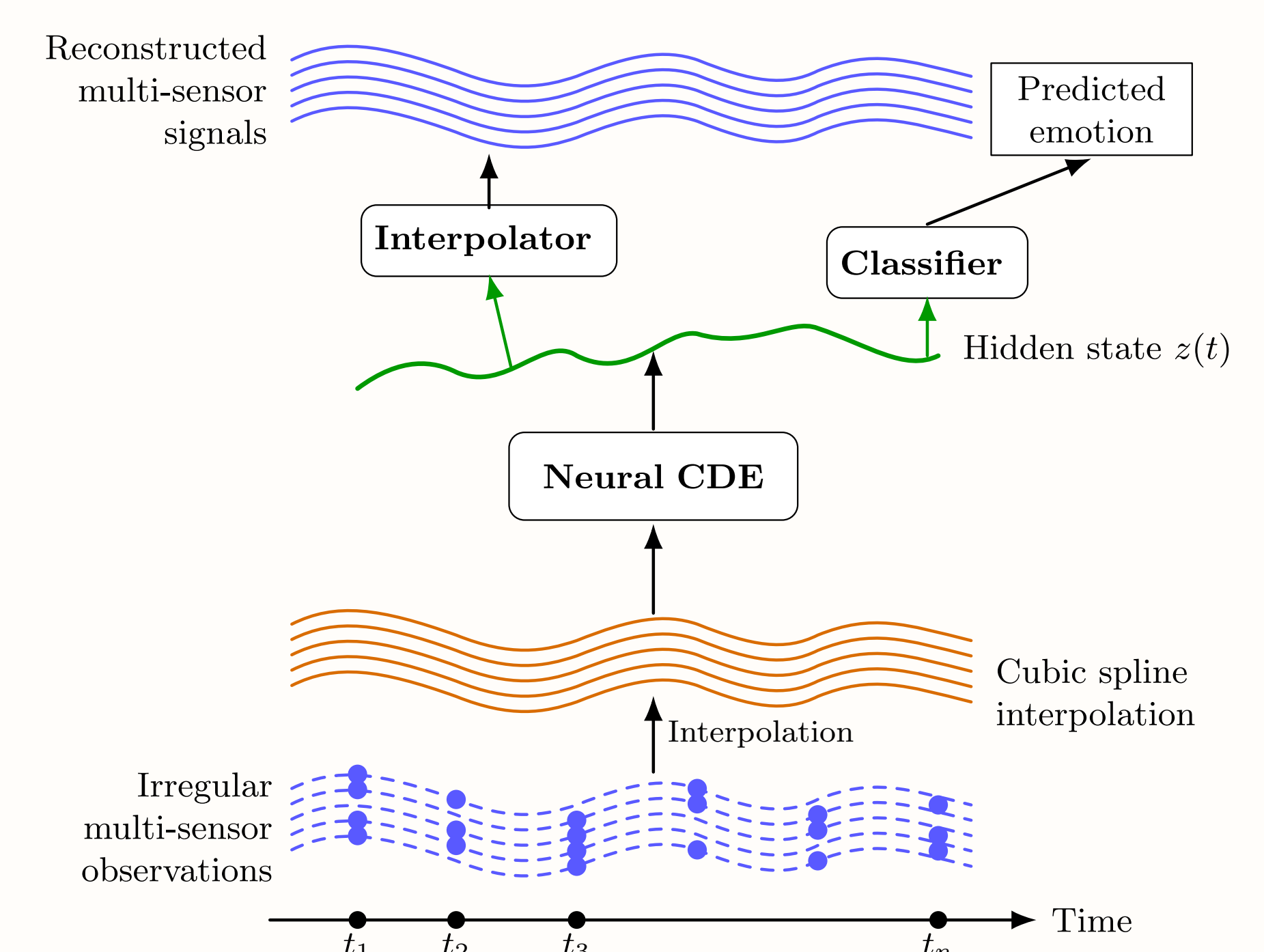


Figure 2: Workflow of the NCDE-based interpolation and classification pipeline.

Results

Method	MSE	Loss	Accuracy (%)
Baseline	0.0962	—	—
NCDE	0.0999	3.617	44.44
U-Net	0.0995	1.792	16.67
TCN	0.0992	1.744	25.00
U-Net + Baseline	0.0962	1.792	16.67
TCN + Baseline	0.1039	1.650	33.33

Table 4: Interpolation and classification results.

Despite NCDE having cubic spline interpolated input, it failed to surpass the baseline, this suggests the problem lies in the data.

Summary and Conclusions

Across all three methodologies, radar sensing shows clear promise for privacy-preserving sentiment recognition, though performance on subtle emotional states remains behind that on simpler, more repetitive behaviors, and behind infrared-based approaches. Feature-based SVM classification proved the most robust and data-efficient approach on the emotion classes studied. Deep learning on raw data generalizes well to unseen behaviors in the unsupervised setting but degrades on fine-grained emotion recognition. Joint interpolation-classification remains constrained by limited data and fails to improve on baseline.

Promising future directions include multimodal fusion of sensors, architectures tailored to point-cloud and irregular-time-series structure, and aligning learned sensor representations with text encoders to enable zero-shot recognition of new behaviors.